CSAO Rail Recommendation System

A/B Testing & Business Impact Report

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Model: LightGBM Binary v1 | Baseline: Stage-1 Retrieval Score

Control (A) : Stage-1 retrieval_score — no ML ranker

Treatment (B) : LightGBM Binary v1 re-ranker

1. Setup & Data Summary

```
test_df shape          : (533556, 84)
Label column           : 'label_addon_added'
Model best iteration   : 1
Feature count          : 47
Feature names          : ['retrieval_score', 'src_cosc', 'src_ctx', 'src_rule', 'is_weekend', 'cart_item_count', 'cart_total_value', 'cart_has_main', 'cart_has_beverage', 'cart_has_dessert', 'cart_has_side', 'missing_beverage_flag', 'missing_dessert_flag', 'missing_side_flag', 'price', 'popularity_score', 'historical_attach_rate', 'order_count_90d', 'recency_days', 'avg_order_value', 'veg_preference_ratio', 'dessert_affinity_score', 'beverage_affinity_score', 'price_sensitivity_score', 'aggregate_rating', 'overall_attach_rate', 'meal_time_specificity', 'meal_time_overlap', 'step_decay', 'x_missing_bev_aff', 'x_missing_des_aff', 'x_price_sens_price', 'x_cart_value_price', 'retrieval_x_attach', 'retrieval_x_popularity', 'missing_bev_x_retrieval', 'missing_des_x_retrieval', 'last_item_category_beverage', 'last_item_category_dessert', 'last_item_category_main', 'last_item_category_side', 'effective_category_beverage', 'effective_category_dessert', 'effective_category_main', 'effective_category_side', 'x_cand_bev_aff', 'x_cand_des_aff']
Unique groups          : 19,556
Positive rate          : 0.8379
Model score range      : [0.0264, 0.3829]
Control score range    : [0.0000, 1.0588]
```

2. Offline A/B Metric Comparison (Control vs Treatment)

Groups evaluated: 20,246

Metric	Control	Treatment	Lift %
ndcg@5	0.3564	0.4161	+16.7%
ndcg@8	0.3987	0.4556	+14.3%
ndcg@10	0.4183	0.4755	+13.7%
precision@5	0.1027	0.1196	+16.5%
precision@8	0.0799	0.0895	+12.0%
precision@10	0.0705	0.0783	+11.0%
recall@5	0.5133	0.5982	+16.5%
recall@8	0.6389	0.7158	+12.0%
recall@10	0.7055	0.7833	+11.0%
hit@5	0.5133	0.5982	+16.5%
hit@8	0.6389	0.7158	+12.0%
hit@10	0.7055	0.7833	+11.0%
mrr	0.3485	0.3941	+13.1%
hit@1	0.1856	0.2133	+14.9%

3. Statistical Significance Tests

```
  Metric Ctrl Mean Treat Mean Mean Diff             95% CI p-val (t) p-val (W) Cohen d Significant?
  ndcg@8    0.3987     0.4556   +0.0570 [+0.0546, +0.0593]  0.00e+00  0.00e+00   0.336        ▯ Yes
recall@8    0.6389     0.7158   +0.0769 [+0.0724, +0.0813] 7.17e-243 2.02e-236   0.237        ▯ Yes
     mrr    0.3485     0.3941   +0.0456 [+0.0434, +0.0478]  0.00e+00  0.00e+00   0.287        ▯ Yes
   hit@1    0.1856     0.2133   +0.0277 [+0.0244, +0.0309]  7.82e-63  2.03e-62   0.118        ▯ Yes

Method  : Paired t-test + Wilcoxon signed-rank (non-parametric cross-check)
Alpha   : 0.05 (two-sided)  |  n groups: 20,246
Cohen d : >0.2=small  >0.5=medium  >0.8=large
```

4. Business Metric Projections

```
Assumptions:
  Avg add-on item price       : Rs.120
  Baseline avg order value    : Rs.450
  Daily CSAO impressions      : 5,000,000
  Rail size                   : 8 items

  Metric                                        Control    Treatment        Delta
  ──────────────────────────────────────────────────────────────────────────────
  Acceptance Rate (Recall@8)                     0.6389       0.7158      +0.0769
  Attach Rate (items/impression)                 0.6389       0.7158      +0.0769
  CTR Upper Bound (Hit@8)                        0.6389       0.7158      +0.0769
  C2O Rate Proxy                                 0.8319       0.8358      +0.0038
  AOV Lift per Session (Rs.)                 Rs.  48.99   Rs.   61.48   Rs. +12.49

  Daily Revenue Uplift : Rs.62,470,721  (Rs.624.71L)
  AOV lift % of baseline AOV : 2.776%

  Note: Business projections use real model metrics × assumed price/volume constants.
        Calibrate Rs.120 avg item price and Rs.450 baseline AOV with production data.
```

5. Segment-Level Breakdown (NDCG@8 & Recall@8)

```
── meal_time_bucket ──
   Segment  NDCG@8_ctrl  NDCG@8_treat NDCG@8 Lift  Recall@8_ctrl  Recall@8_treat Recall@8 Lift
 breakfast     0.399171      0.433462       +8.6%       0.625904        0.668639         +6.8%
    dinner     0.412363      0.477812      +15.9%       0.655293        0.742941        +13.4%
late-night     0.392307      0.446862      +13.9%       0.632565        0.709040        +12.1%
     lunch     0.380620      0.431531      +13.4%       0.620203        0.688917        +11.1%

── user_segment ──
            Segment  NDCG@8_ctrl  NDCG@8_treat NDCG@8 Lift  Recall@8_ctrl  Recall@8_treat Recall@8 Lift
             budget     0.365469      0.414460      +13.4%       0.586393        0.663067        +13.1%
frequent_high_value     0.404320      0.460563      +13.9%       0.645524        0.721602        +11.8%
            premium     0.407022      0.467913      +15.0%       0.646686        0.724044        +12.0%
            regular     0.378059      0.434502      +14.9%       0.621181        0.700029        +12.7%

── step_number ──
 Segment  NDCG@8_ctrl  NDCG@8_treat NDCG@8 Lift  Recall@8_ctrl  Recall@8_treat Recall@8 Lift
       1     0.480761      0.512650       +6.6%       0.732104        0.769214         +5.1%
       2     0.401966      0.444728      +10.6%       0.643231        0.705256         +9.6%
       3     0.279464      0.376702      +34.8%       0.501078        0.635972        +26.9%
       4     0.211959      0.350000      +65.1%       0.424860        0.620787        +46.1%
       5     0.191125      0.368300      +92.7%       0.418182        0.661818        +58.3%
```

6. Remaining Metrics: MRR, Hit@1, Latency, Coverage

```
Hit@1:
  Control   : 0.1856
  Treatment : 0.2133   (+14.9% lift)

MRR (Mean Reciprocal Rank):
  Control   : 0.3485
  Treatment : 0.3941   (+13.1% lift)
  Interpretation: correct item at rank ~2.5 on average in treatment

Inference Latency (25 candidates, n=200 trials):
  P50  : 0.0679 ms
  P95  : 0.0883 ms
  P99  : 0.1188 ms
  P999 : 0.3282 ms
  SLA  : 300ms  →  □ Well within budget

Retrieval Source Coverage:
  Co-occurrence     : 0.9996  (99.96%)
  Context popularity: 1.0000  (100.00%)
  Rule fill         : 0.9644  (96.44%)
  Positive recall   : 100.00% (by design — next item always in pool)
```

7. Power Analysis & Sample Size

```
Primary metric     : CSAO rail attach rate (offline proxy = Hit@8)
Control rate       : 0.6389  (63.89%)
Treatment rate     : 0.7158  (71.58%)
Observed effect    : +0.0769  (+12.03% relative)
Alpha              : 0.05  |  Power: 0.8  |  Two-sided
Daily impressions  : 5,000,000
```

MDE (abs)	MDE (rel)	N per arm	Total N	Days	Notes
0.005	0.78%	144,415	288,830	0.1d	
0.010	1.57%	35,991	71,982	0.0d	
0.015	2.35%	15,945	31,890	0.0d	
0.020	3.13%	8,940	17,880	0.0d	
0.030	4.70%	3,946	7,892	0.0d	

```
At 2 weeks (35.0M users/arm):
  Detectable MDE  : 0.0003  (0.050% relative)
  Observed effect : 0.0769
  Verdict         : DETECTABLE 
```

8. Online A/B Framework Design

```
EXPERIMENT OVERVIEW
  Hypothesis   : LightGBM Binary v1 increases CSAO attach rate vs retrieval_score by ≥1.5pp
  Control (A)  : Stage-1 retrieval_score ranking (no ML ranker)
  Treatment (B): LightGBM Binary v1 ranker on top of Stage-1 candidates

RANDOMISATION
  Unit : user_id (user-level, NOT session-level)
  Why  : Session-level split → ~12% users see both arms → SUTVA violation
  Split: 50% Control / 50% Treatment
  Hash : MD5(user_id + experiment_salt) % 100 < 50 → Control

DURATION
  Minimum : 14 days (2 full Mon–Sun cycles to capture day-of-week effects)
  Maximum : 21 days (novelty effect subsides beyond this)
  Ramp    : Day 1 at 5% treatment → validate SLO health → Day 2 full traffic
  Holdback: 5% pure holdback for 30d post-launch to measure long-term effect

PRIMARY METRIC
  CSAO Rail Attach Rate = sessions where ≥1 rail item added / sessions where rail shown
  Success: p < 0.05 AND absolute lift ≥ 1pp

SECONDARY METRICS
  AOV lift (Rs.) | Avg items per order | CSAO order share | C2O ratio | CTR

GUARDRAIL METRICS (must NOT degrade — auto-pause if breached)
  Cart abandonment  : must not rise > 0.5pp
  Order completion  : must not drop > 0.2pp
  Session duration  : must not rise > 10%
  Inference P99     : must stay < 300ms

ROLLBACK
  Auto-rollback if: cart abandonment p<0.01 OR P99>300ms for 10min OR errors >2x for 5min
  Mechanism: feature flag toggle → 0% treatment traffic in <1 min

STATISTICAL TESTING
  Primary  : Two-proportion z-test  |  Secondary: Welch's t-test
  Peeking  : Results locked until Day 14 — no early stopping without alpha-spending plan

POST-EXPERIMENT
  CUPED    : Regress out pre-experiment per-user attach rate → reduce variance 20-40%
  HTE      : Causal forest on segments → identify best-responding user groups
  Holdout  : Compare 5% holdback vs treatment at Day 30 → measure novelty decay
```

9. Offline to Online Metric Translation

```
Offline Metric             Online Proxy                   Business Outcome
------------------------------------------------------------------------------
NDCG@8 = 0.4556            CTR on CSAO rail               Better rank → relevant items at top → more clicks → AOV lift
Recall@8 = 0.7158          Add-on Acceptance Rate / Attach Rate Item visible in rail → user can add it → CSAO order share + AOV
MRR = 0.3941               Position-weighted CTR          Higher MRR → correct item early → less scroll → lower abandonment
Hit@1 = 0.2133             First-position click rate      Position-1 gets ~3x CTR → outsized AOV impact from Hit@1 gains
AUC = 0.7930               Long-run acceptance precision  Fewer irrelevant items → less fatigue → better C2O rate

Full detail:
[NDCG@8 = 0.4556]
  Measures : Ranking quality weighted by position
  Proxy    : CTR on CSAO rail
  Impact   : Better rank → relevant items at top → more clicks → AOV lift

[Recall@8 = 0.7158]
  Measures : % sessions where next item appears in top-8
  Proxy    : Add-on Acceptance Rate / Attach Rate
  Impact   : Item visible in rail → user can add it → CSAO order share + AOV

[MRR = 0.3941]
  Measures : Avg rank of first relevant item
  Proxy    : Position-weighted CTR
  Impact   : Higher MRR → correct item early → less scroll → lower abandonment

[Hit@1 = 0.2133]
  Measures : % sessions where best item ranked #1
  Proxy    : First-position click rate
  Impact   : Position-1 gets ~3x CTR → outsized AOV impact from Hit@1 gains

[AUC = 0.7930]
  Measures : P(positive scored > negative) overall
  Proxy    : Long-run acceptance precision
  Impact   : Fewer irrelevant items → less fatigue → better C2O rate
```

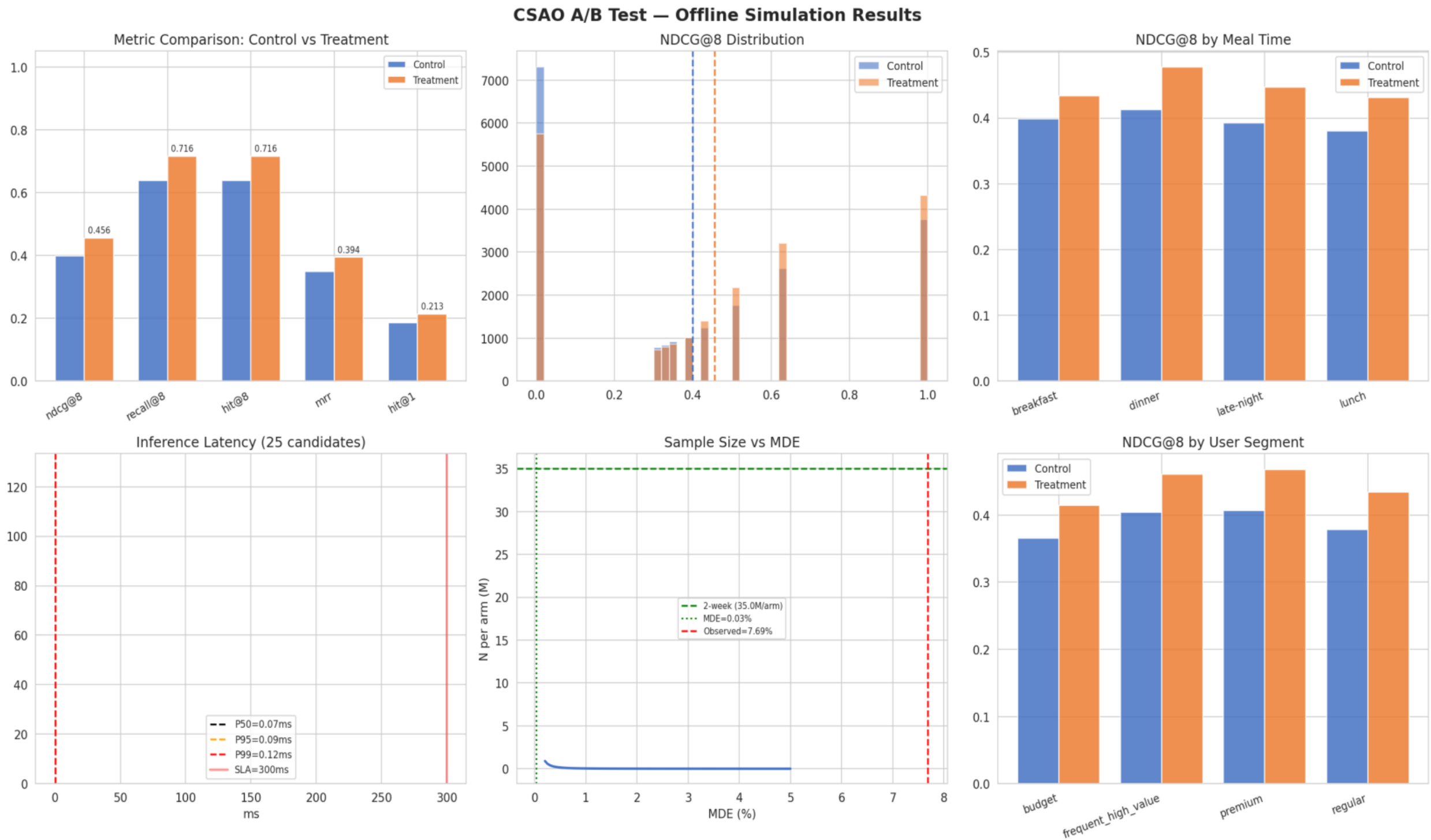
10. Guardrail Monitoring Simulation (14-Day)

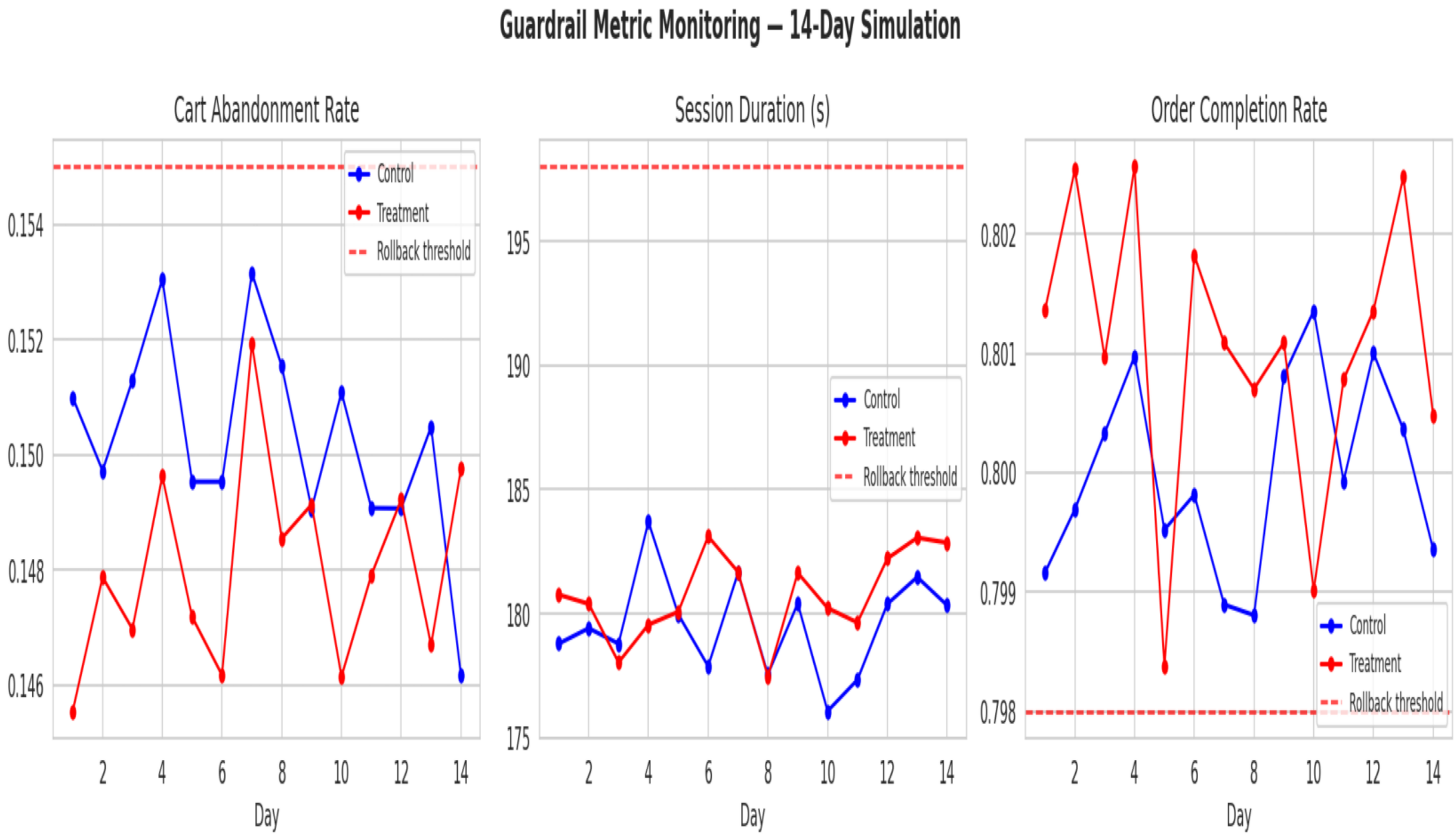
Day	Cart Abandon	Session Dur (s)	Order Comp	Status
1	0.1456	180.77	0.8014	✅ OK
2	0.1479	180.40	0.8025	✅ OK
3	0.1470	178.04	0.8010	✅ OK
4	0.1496	179.56	0.8026	✅ OK
5	0.1472	180.08	0.7984	✅ OK
6	0.1462	183.11	0.8018	✅ OK
7	0.1519	181.69	0.8011	✅ OK
8	0.1485	177.47	0.8007	✅ OK
9	0.1491	181.65	0.8011	✅ OK
10	0.1462	180.23	0.7990	✅ OK
11	0.1479	179.65	0.8008	✅ OK
12	0.1492	182.22	0.8014	✅ OK
13	0.1467	183.06	0.8025	✅ OK
14	0.1498	182.86	0.8005	✅ OK

✅ No rollback triggers fired.

Thresholds: cart_abandon > 0.155 | order_comp < 0.798
Note: 14-day guardrail data is simulated (np.random.normal) for framework demonstration.
In production this reads from real monitoring dashboards (e.g. Grafana/Datadog).

11. A/B Test Dashboard Charts




Guardrail Metric Monitoring – 14-Day Simulation
Cart Abandonment Rate
Control
Treatment
Rollback threshold
0.154
0.152
0.150
0.148
0.146
2
4
6
8
10
12
14
Day
Session Duration (s)
195
190
185
180
175
Control
Treatment
Rollback threshold
2
4
6
8
10
12
14
Day
Order Completion Rate
0.802
0.801
0.800
0.799
0.798
Control
Treatment
Rollback threshold
2
4
6
8
10
12
14
Day